

Continuous Front-End Signal Flow Simulation in Nuclear Instrumentation Using the Lorenzetti Framework

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Abstract—Particle detector simulators, such as the Lorenzetti framework, are essential tools for studying and designing new acquisition architectures for nuclear instrumentation. Generally, these simulators encompass the entire readout chain, from pulse digitization to sample storage. Although well-established, this event-driven simulation philosophy presents limitations when studying first-level (L1) trigger algorithms, which operate heavily on continuous data streams. Currently, it is not possible to use standard simulators to effectively analyze this type of real-time processing. To bridge this gap, this work proposes a reinterpretation of Minimum Bias simulation data, transforming it into a continuous sequence of collisions. The proposed method enables the generation of continuous data in a free-running regime, providing a robust framework for the design, testing, and development of real-time calorimetry data processing systems, specifically targeting low-level trigger algorithms within the Lorenzetti environment.

Index Terms—High-Luminosity LHC, Calorimetry, Pile-up Simulation, Free-Running Readout, Low-Level Trigger, Data Processing.

I. INTRODUCTION

Particle detectors operating in high-energy colliders are highly complex instruments, designed to identify and measure the fundamental properties of the subatomic particles generated following collisions [13]. Among the subsystems that comprise these detectors, calorimeters are of central importance. In addition to determining the total deposited energy, their high spatial segmentation allows the identity of incident particles to be inferred through the morphological profile of the radiation shower generated within the material [12].

The transformation of these particle showers into useful data occurs through the conversion of the energy deposited

in the active material into electrical pulses, which are subsequently digitised by analogue-to-digital converters (ADCs) [11]. In modern experiments, such as the *Large Hadron Collider* (LHC), *trigger* systems form a fundamental part of data acquisition [6]. By processing data from a large number of readout channels uninterruptedly, these systems must make microsecond decisions regarding which events should be stored, thereby enabling subsequent reconstructions and physical analyses [8].

The major challenge for the next generation of experiments, the *High-Luminosity LHC* (HL-LHC), operating at a centre-of-mass energy of 14 TeV, is the increase in the average number of collisions per bunch crossing [9]. This scenario exacerbates the effects of signal *pile-up* [4] and *crosstalk* [11], where the same readout channel processes data from temporally adjacent collisions, degrading the estimation of the amplitude and phase of the signal of interest. To mitigate these effects at the *trigger* system level, *machine learning* techniques have been increasingly employed [1], [2].

In the context of the need to generate data that simulate particle collisions, the Lorenzetti framework stands out [2]. Conceived within the scope of the ATLAS collaboration, the framework is a general-purpose simulator focused on calorimetry, distinguishing itself by offering a modular and flexible environment. It provides mechanisms to simulate the geometry of the calorimeters and the underlying particle collisions, aiming to support the development of signal reconstruction methods and *trigger* algorithms.

However, like detector simulation *frameworks* in particle colliders in general, Lorenzetti was designed for the simulation of windowed data, that is, a fixed number of temporal samples per readout channel for each collision. In this approach, the signal of interest is processed as a predefined temporal

snapshot, emulating the behaviour of systems that already operate under a *trigger* decision. This characteristic results in the loss of the historical continuity of sequentially acquired data, which precludes the use of these tools for the study of signal processing in a *free-running* regime. Such analyses are fundamental for evaluating, for example, the efficiency of the first-level *trigger* and the behaviour of noise in continuous data streams.

This work proposes an alteration in the method of collecting and interpreting *minimum bias* simulation data from the Lorenzetti *framework*, in order to add this further functionality to the *framework*: the possibility of analysing continuous data. With this expansion, Lorenzetti becomes capable of supporting more complex studies on the uninterrupted flow of signals, bringing the simulation closer to the real operating conditions of next-generation data acquisition systems.

II. THE LORENZETTI FRAMEWORK

The *Lorenzetti* framework is an advanced computational system for data simulation and reconstruction, developed to optimise physics performance with low-level calorimetry data within the context of the ATLAS experiment [2]. Its architecture is highly modular, employing the *Gaudi* library — an adaptation of *Gaudi* [14] — for orchestration, and *Geant4* [7] as the particle propagation engine. The simulation lifecycle is managed by the *ComponentAccumulator* module, with data persisted in the ROOT tree format [5].

To emulate the high-luminosity conditions of the HL-LHC, the framework incorporates an overlay module responsible for integrating background noise (*pile-up*). The signal of interest is superimposed onto a sampling of *Minimum Bias* events, governed by a Poisson distribution based on the μ parameter, which defines the average number of interactions per bunch crossing.

At the final stage of the simulation, the system extracts morphological features such as *clusters* and concentric rings (*CaloRings*), exporting the generated data in the structured format of ROOT trees. However, the native structure of these trees stores the detector response in the form of hierarchical and variable-sized vector arrays, a format that is incompatible with the requirements for the design and development of real-time signal processing systems.

III. CONTINUOUS MINIMUM BIAS SIMULATION

To model the generation of continuous data, we use *Minimum Bias* simulations, which represent generic proton-proton collisions without specific process selection. The generation of this background noise is performed with the Monte Carlo generator *Pythia 8* [3], configured for 14 TeV collisions. The physics engine simulates inelastic low-momentum transfer processes through the activation of the `SoftQCD:all` flag. *Pythia 8* calculates the kinematics and nature of the resulting particles, whose data are persisted in ROOT format. This final file serves as the standardised input for *Geant4*, which will perform the simulation of the transport of these particles through the detector [2].

Continuing the *pipeline*, the simulation of the interaction of these particles with matter is executed by the *Geant4* engine. This process is fundamental for converting mathematical trajectories into physical signals, calculating, step by step, the propagation through the complex geometries of the detector, including the liquid argon electromagnetic and scintillator hadronic calorimeters. During propagation, the system models phenomena such as ionisation losses and the development of radiation showers, cataloguing the energy deposited in each sensitive volume.

To convert these independent events into a continuous data stream, the algorithm mimics the accelerator dynamics through the $\langle\mu\rangle$ parameter, which defines the average number of collisions per bunch crossing (*bunch-crossing*). The procedure is based on a statistical draw via a Poisson distribution, using the value of $\langle\mu\rangle$ to determine the exact number of collisions (N) to be superimposed at each instant, see Figure 1. Once N is defined, the deposited energies of n consecutive *Minimum Bias* events are summed, consolidating the total detector response for that specific crossing. This cycle of drawing and aggregation is repeated sequentially, incrementing the temporal sample until the entire database is processed, resulting in a realistic representation of the pile-up effect (*pile-up*) observed at the LHC.

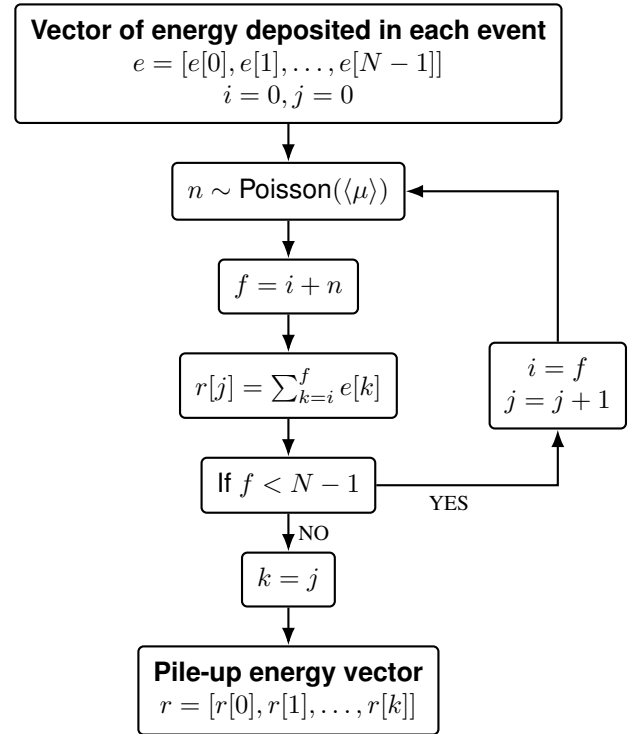


Figure 1: Data synthesis logic for background noise. The algorithm iterates over the energy deposition data, grouping and summing consecutive *Minimum Bias* events. The quantity of aggregated events at each time step is defined by a stochastic draw ($\text{Poisson}(\langle\mu\rangle)$), accurately mimicking signal superposition within the accelerator environment.

IV. SIMULATION VALIDATION METRICS

The calorimetry system on which this work was based is inspired in the ATLAS calorimeter whose segmentation is shown in Figure 2 [15]. Note that for each detector (electromagnetic or hadronic calorimeter) and layer, there is an associated pseudorapidity (η) region. It is expected that the energy deposition rate per event will show an increase in cells with higher $|\eta|$ values, as these regions cover smaller angles relative to the beam axis, where particle flux and energy density are naturally higher. Additionally, this deposition profile should vary according to the depth of the layers, being influenced by the distance from the collision point and the development of radiation showers.

To validate the simulator's integrity in generating *Minimum Bias* events, two fundamental analyses will be conducted. The first consists of estimating cell occupancy — defined as the ratio of events with energy deposition to the total number of simulated events — segmented by $|\eta|$ and calorimeter layer. The second analysis is based on the integration of energy in the $\eta \times \phi$ plane; it is expected that the integrated energy will grow proportionally with $|\eta|$ due to collision kinematics, but remain statistically uniform with respect to the azimuthal angle ϕ , reflecting the cylindrical symmetry of the experiment.

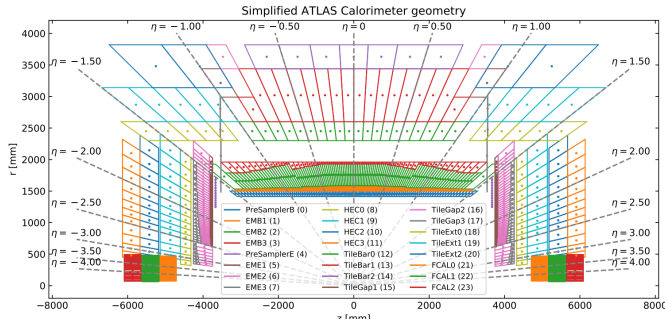


Figure 2: Geometry of the calorimetry system used.

A. Occupancy Rate

In the context of calorimetry, the occupancy rate is defined as the probability of a specific detector cell registering an energy deposit during a single bunch-crossing event. Mathematically, the percentage occupancy (OC) can be calculated using the following equation (1):

$$OC = \frac{N_{edep}}{2 \cdot N_{evt} \cdot N_{mod}} \cdot 100, \quad (1)$$

where N_{edep} corresponds to the number of events in which energy deposition occurred, N_{evt} is the total number of events considered, and N_{mod} is the number of modules present in the ϕ segmentation. It is necessary to normalise by this number, as the data will be integrated in ϕ to increase statistics. The second and third layers of the electromagnetic calorimeter have 256 modules, while the other detectors have 64. The factor of two is present in the denominator because the analysis adopts $|\eta|$; therefore, an average of the contribution from the positive and negative parts of the pseudorapidity is performed.

To ensure statistical convergence, this *pipeline* processed a database of 3,000 *Minimum Bias* events at 14 TeV. The vector results obtained allowed for the precise delineation of the activation probability throughout the detector geometry.

B. Energy Topological Mapping

The validation of the stochastic model is consolidated by correlating the spatial location of the radiation with the magnitude of the energy depositions. While the one-dimensional occupancy profiles validate the activation probability, the topological mapping of energy in η and ϕ demonstrates the simulator's efficiency in correctly characterising energy deposition within the calorimetry system's readout channels. This analysis ensures that the Poisson distribution distributes the *Minimum Bias* energy while strictly respecting the phenomenology and physical characteristics of the detector.

In the electromagnetic layers, the energy distribution should highlight azimuthal symmetry and accurately identify intrinsic instrumentation discontinuities. Similarly, the behaviour of the hadronic calorimeter must confirm the effectiveness of the prior electromagnetic shielding and the scattering kinematics in high-energy colliders.

Complementary to positional precision, the characterisation of signal magnitude should reveal the signature of soft inelastic interactions. In this case, the energy spectrum shows a predominant concentration at low amplitudes, with an exponential decay towards rare high-magnitude events. Faithfully modelling this behaviour is critical for training *trigger* systems, as stochastic energy fluctuations are the main sources of false triggers (*fake triggers*).

V. RESULTS AND DISCUSSIONS

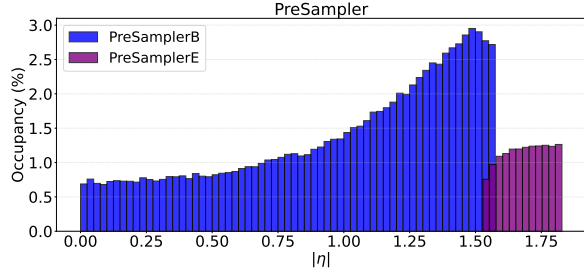
The presentation of the results begins with the statistical quantification of the *Minimum Bias* simulation integrated into the Lorenzetti *framework*. This fundamental step seeks to validate the data generated by the simulator through two complementary fronts: the measurement of cell occupancy and the topological mapping of energy deposition. Through these metrics, it is possible to characterise the behaviour of the noise and verify whether the simulated response reproduces the expected patterns of spatial distribution and energy density prior to the application of the continuous data generation algorithm.

A. Electromagnetic Calorimeter Occupancy Profile

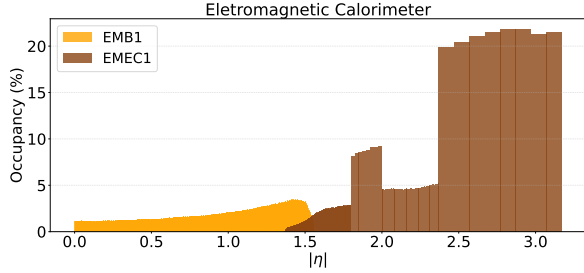
The longitudinal characterisation of the EM calorimeter quantifies the evolution and absorption of the particle showers through the liquid argon and lead layers. The primary interaction in the *Presampler* (Figure 3a) presents a stable activation rate in the central region of the barrel. Physically, this attests that the simulator accurately samples the charged particles before the full development of the shower, serving as a probe for energy loss corrections in passive materials.

The transition to EM Layer 1 (Figure 3b), characterised by the extremely high granularity of its micro-strips, reveals a drastic change in behaviour: the occupancy in the barrel

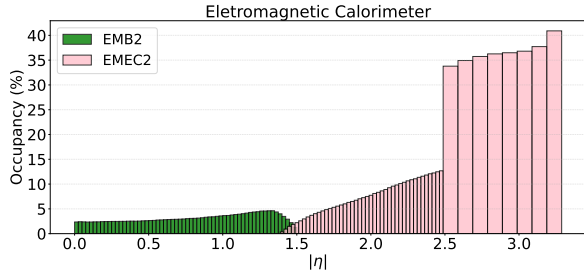
decreases, while the forward partitions (*Endcaps*) record the first significant occupancy peaks at high pseudorapidity.



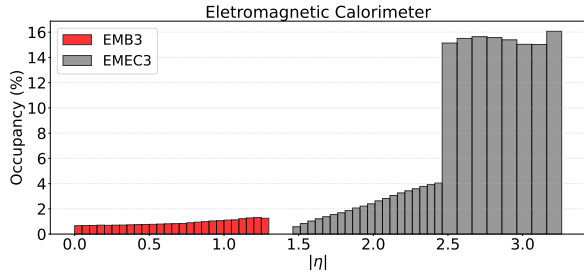
(a) *Presampler*: stability in the central region.



(b) EM Layer 1: occupancy peaks in the *Endcaps*.



(c) EM Layer 2: highest occupancy levels.



(d) EM Layer 3: decay of occupancy levels

Figure 3: Initial stages of electromagnetic shower development.

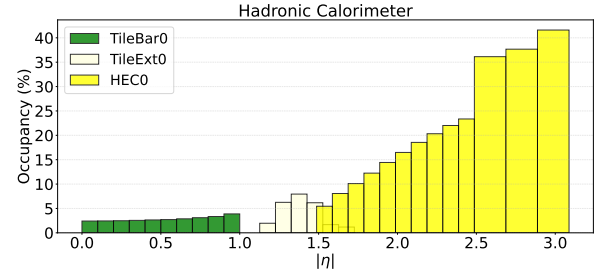
The peak of energy deposition, known as the *shower maximum*, occurs in EM Layer 2 (Figure 3c). The emulator captures with surgical precision the mechanical discontinuity at $|\eta| \approx 1.4$. This abrupt drop in occupancy naturally registers the *Crack Region* (the mechanical transition space between the barrel and the endcaps), proving the absence of distortions in the data extraction. The fact that the simulator registers

this void proves its geometric and granular fidelity. The rapid signal decay in EM Layer 3 (Figure 3d) attests to the almost total absorption of the electromagnetic component of the incident radiation.

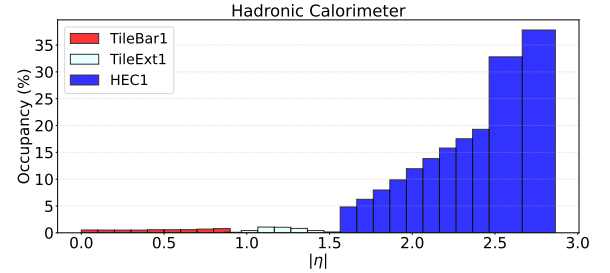
B. Hadronic Calorimeter Occupancy Profile

The response of the hadronic calorimeter (Figure 4) exhibits the expected and severe geometric asymmetry dictated by the kinematics of proton-proton collisions. In the central region ($|\eta| < 1.5$), the occupancy rate of the calorimeter scintillators approaches zero for the second and third layers. This almost absolute suppression proves the effectiveness of the shielding provided by the preceding electromagnetic layers, which absorb the soft *Minimum Bias* noise.

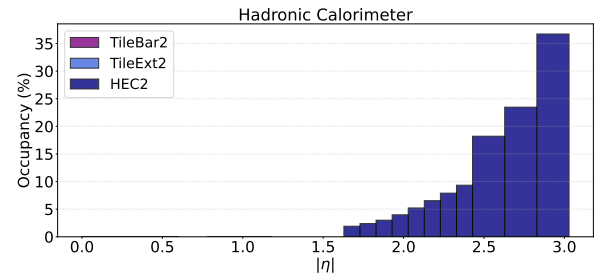
In contrast, the forward partitions (*HEC*) present a significant increase in occupancy, exceeding 4%. This collimated flux of secondary particles around the beam pipe is accurately captured, certifying that *trigger* algorithms trained on these data will be prepared for the high-luminosity regime.



(a) Hadronic Layer 1: Contrast between barrel and Endcap.



(b) Hadronic Layer 2: Intermediate response.



(c) Hadronic Layer 3: Final containment of the radiation.

Figure 4: Hadronic profiles quantifying the shielding in the central barrel and the high occupancy in the endcaps.

C. Topological Mapping and Energy Spectrum

Figure 5 displays the topological mapping of the energy integral corresponding to the PreSamplerB. The uniform distribution in ϕ , as well as the increase in energy as a function of η , attest to the correct behaviour of the simulator. Furthermore, Figure 6 highlights the exponential nature of the energy distribution for layer EMB2, whose signature is in accordance with what is expected for low transverse momentum particles originating from *Minimum Bias* events.

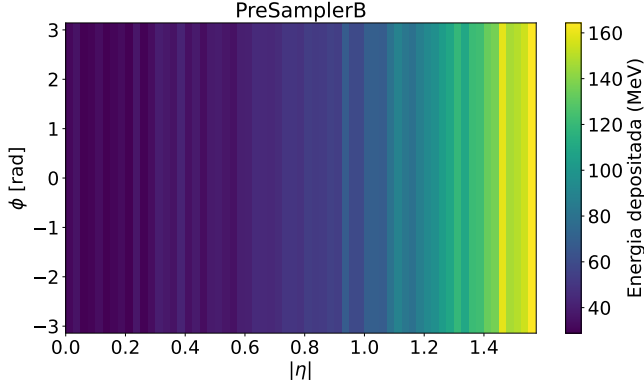


Figure 5: Energy map for the PreSamplerB layer

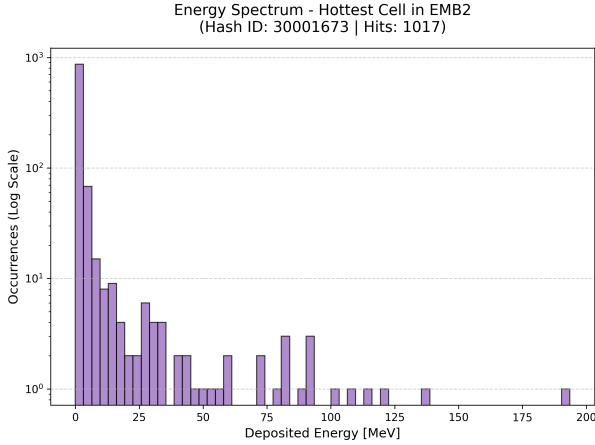


Figure 6: Energy distribution of the PreSamplerB.

D. Synthesis of the Continuous Readout Signal

The final stage of the model consists of extracting the temporal signature resulting from the collision pile-up and converting it into the digitised format read by the detector's instrumentation. The first step of this consolidation is obtaining the *Truth* sequence of pure physical energy. From an expanded set of three thousand simulated events, configured for a high-luminosity scenario ($\mu = 50$), a representative temporal snapshot containing just over sixty bunch crossings was generated. Figure 7 presents this *Truth* signal, which purely reflects the energy fluctuations of the inelastic interactions.

Physical Pile-up (Truth Signal), $\mu = 50$

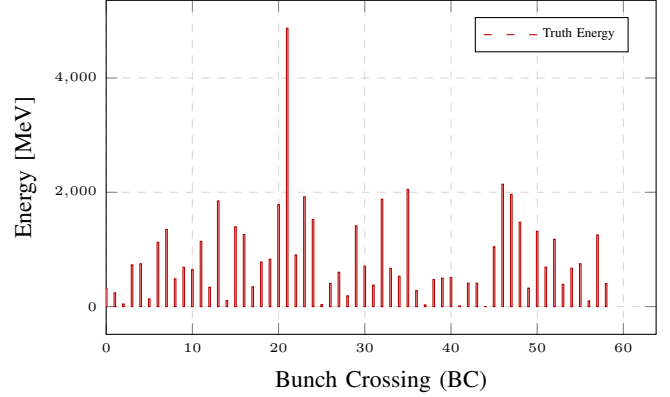


Figure 7: Physical energy deposition per Bunch Crossing (BC) modelled in the Hadronic Layer 3 with $\mu = 50$

To finalise the simulation and obtain the real signal that is seen by the electronic *front-end* immediately after being read by the analogue-to-digital converter (ADC), this same *Truth* sequence ($\mu = 50$) is taken and a discrete convolution is performed with the detector's characteristic pulse. The application of this filter characterises the temporal spreading of the energy imposed by the analogue response of the calorimeter's electronics. The result of this convolution is shown in Figure 8, which illustrates the final digitised electronic signal. This signature, marked by severe superposition and constant baseline fluctuation, represents exactly the continuous readout expected by the acquisition system operating under extreme *pile-up* conditions.

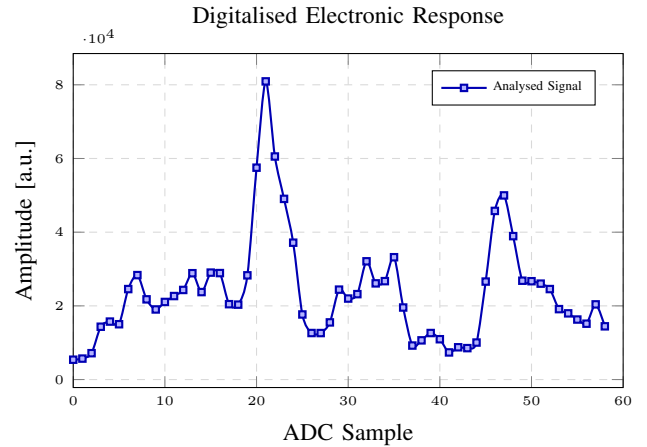


Figure 8: Output signal after convolution with the calorimeter electronic response modelling.

To evaluate the framework under low pile-up conditions, a scenario with $\mu = 10$ was simulated. Compared to the nominal $\mu = 50$, the *Truth* signal (Figure 9) exhibits significantly lower baseline and peak energies. The convolved electronic signal (Figure 10) clearly resolves individual pulses due to

the reduced overlap. This confirms the framework generates reliable continuous data across varying luminosity regimes.

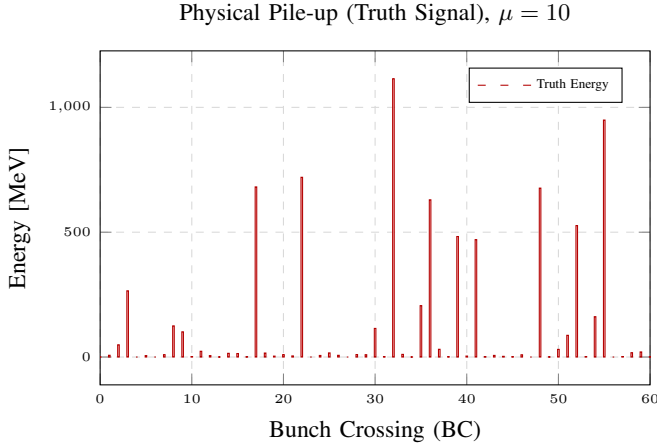


Figure 9: Physical energy deposition per Bunch Crossing (BC) modelled in the Hadronic Layer 3 with $\mu = 10$.

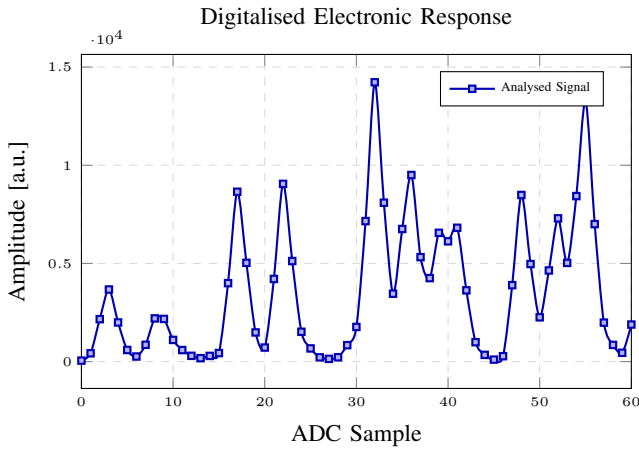


Figure 10: Output signal after convolution with the calorimeter electronic response modelling.

VI. CONCLUSION

A continuous data simulation method for nuclear instrumentation was successfully implemented and tested. For this purpose, the Lorenzetti *framework* was utilised, adding this further simulation functionality to the tool and making it even more comprehensive.

To quantify whether the simulation represents realistic data, the Lorenzetti occupancy rate as a function of pseudorapidity ($|\eta|$) and calorimeter layer was investigated—an analysis that had not yet been validated by the collaboration. The extraction of occupancy statistics and the topological mapping of energy deposition demonstrated that the *Minimum Bias* simulation generated within the *framework* is highly reliable.

The proposed algorithm for the population of bunch crossings using the Poisson distribution for different values of $\langle\mu\rangle$

can already be used by the collaboration for the study and analysis of real-time data processing methods.

As future work, it is planned to utilise the additional *Time of Flight* (ToF) information of the particles, provided by Geant4, to perform convolutions with the impulse response by applying phase shifts. This will allow for the synthesis of a simulated *front-end* signal with even greater precision.

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